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Artificial neural network-based pharmacogenomic algorithm for warfarin dose optimization

Aim: To develop more precise pharmacogenomic algorithm for prediction of safe and effective dose of warfarin. **Materials & methods:** An artificial neural network (ANN) algorithm was developed by using age, gender, BMI, plasma vitamin K levels, thyroid status and ten genetic variables as the inputs and therapeutic warfarin dose as the output. Hyperbolic tangent function was used to build an ANN architecture. **Results:** This model explained 93.5% variability in warfarin dosing and predicted warfarin dose accurately in 74.5% patients whose international normalized ratio (INR) was less than 2.0 and in 83.3% patients whose INR was more than 3.5. This algorithm reduced the out-of-range INRs (odds ratio [OR]: 0.49; 95% CI: 0.30–0.79; $p = 0.003$), the rate of adverse drug reactions (OR: 0.00; 95% CI: 0.00–1.21; $p = 0.06$) and time to reach first therapeutic INR (OR: 6.73; 95% CI: 2.17–22.31; $p < 0.0001$). This algorithm was found to be applicable in both euthyroid and hypothyroid status. S-warfarin/7-hydroxywarfarin ratio was found to increase in subjects with *CYP2C9*2* and *CYP2C9*3* justifying the warfarin sensitivity attributed to these variants. **Conclusion:** An application of ANN for warfarin dosing improves predictability and provides safe and effective dosing.

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Keywords: artificial neural networks • *CYP2C9*2* • *CYP2C9*3* • *CYP2C9*8* • warfarin/7-OH warfarin ratio

Background

Warfarin is one of the most commonly prescribed oral anticoagulant with wide inter-individual variability in the dose requirements and narrow therapeutic index [1]. The commercially available warfarin is a racemic mixture of *R*- and *S*-enantiomers with *S*-warfarin being three- to five-times more potent than *R*-warfarin [2]. Warfarin inhibits VKORC1, thus interfering with vitamin K-dependent activation of coagulation factors of extrinsic pathway [3]. Cytochrome P_{450} (CYP)2C9 catalyzes 7-hydroxylation of *S*-warfarin [4]. The poor metabolizer phenotype of CYP2C9, which is attributed to *CYP2C9*2* (rs1799853) and *CYP2C9*3* (rs1057910) polymorphisms, results in warfarin sensitivity and increases

the risk for bleeding events during warfarin therapy [5–7]. *CYP2C9*8* (rs7900194) polymorphism was shown to exhibit lower *S*-warfarin clearance [8]. This variant allele was reported to be prevalent in African-Americans [9] while *CYP2C9*13* (rs72558187) allele was observed in Han Chinese population with a frequency of 0.7% [10]. Promoter polymorphism in VKORC1, in other words, -1639 G>A was also found to be associated with warfarin sensitivity [11]. Another CYP enzyme, in other words, CYP4F2 that hydroxylates vitamin K phytyl side chain, also was explored for possible influence on the warfarin dosing. The rs2108622 *CYP4F2* polymorphism was shown to increase individual requirement of warfarin [12]. *GGCX* rs11676382 was shown

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to be associated with 27% lower-dose requirement of warfarin [13].

Several pharmacogenomic algorithms were developed to optimize warfarin dose based on the demographic and genetic factors [14–19]. These algorithms were based on multiple linear regression models. Recently, we have performed comparative analysis of multiple linear regression model with multiple polynomial regression model and found the additional benefits of multiple polynomial regression models [20]. With increasing number of genetic variants reported to influence pharmacokinetic and pharmacodynamic pathways of warfarin, other parameters such as thyroid status [21], plasma vitamin K profile [22] exerting their impact on warfarin dose requirement, there is an increased demand for algorithms that can incorporate many input variables and can explain more variability in the warfarin dosing. Artificial neural networks (ANNs) are apt for such applications as: first, they can incorporate large number of inputs and large number of outputs; second, they are dynamic and can be upgraded with new data sets periodically; third, they can explore the nonlinear relationships as well and hence suited to study haplotype–haplotype associations. In the current study, we have aimed to develop a precise pharmacogenomic algorithm for warfarin using ANN model and to test its clinical utility in an independent data set. Furthermore, we have elucidated the impact of *CYP2C9**2 and *CYP2C9**3 variants on the enantiomer-specific metabolism of warfarin to substantiate these results.

Materials & methods

Recruitment of subjects

A total of 240 patients who were prescribed to take warfarin were divided into two groups based on their two concurrent international normalized ratio (INR) values: first, derivation cohort (n = 157): INR between 2.0 and 3.5 (mean age: 38.18 ± 15.47 years); second, validation cohort (n = 83): INR less than 2.0 and INR more than 3.5 (mean age: 36.98 ± 17.37 years). In derivation cohort, 153 patients (82 men and 71 women) underwent valve replacement therapy, two patients had deep vein thrombosis and two patients were referred for atrial fibrillation; all 83 patients (46 men and 37 women) in validation cohort underwent valve replacement therapy. The rationale to include patients with out-of-range INRs was to assess whether the developed model can predict higher dose in INR less than 2.0 and lesser dose in INR more than 3.5 so as to stabilize their maintenance dose. Patients were prescribed to take warfarin at 8 a.m. after half an hour of breakfast and blood samples were collected after 3 h to determine the steady-state concentrations of warfarin enantiomers and 7-hydroxy warfarin. The study protocol was approved by the Institutional

Ethical Committee of Nizam's Institute of Medical Sciences, Hyderabad, India (EC/NIMS/1170/2010). Informed consent was obtained from all the patients. A standard questionnaire was developed to record the demographic, medical history and adverse drug reactions in each patient.

Biochemical analysis

Plasma thyroxine (T4) and thyroid-stimulating hormone (TSH) were measured in all the patients by using commercially available ELISA kits (Weldon Biotech Pvt Ltd, New Delhi, India; and Calbiotech Inc, CA, USA) respectively). The inter-assay and intra-assay precisions were 4.0 and 4.6%, respectively. These two parameters coupled with clinical details were used to diagnose hypothyroidism.

Genetic analysis

Whole blood samples were collected in EDTA. The plasma was separated as early as possible following centrifugation and stored at -80°C until the analysis. The buffycoat was used for genomic DNA isolation using the phenol–chloroform extraction method. PCR-restriction fragment length polymorphism (PCR-RFLP) analyses were carried out for detecting *CYP2C9**2, *CYP2C9**3, *CYP2C9**8, *CYP2C9**13, *VKORC1**3, *VKORC1**4, *VKORC1* -1639 G>A, *VKORC1* D36Y, *CYP4F2* V433M and *GGCX* G8016A genetic variants as described in Table 1 [18,19].

Haplotype construction

The data on the derivation cohort were segregated into warfarin sensitive (<28 mg/week), intermediate requirement (28–49 mg/week) and warfarin resistant (>49 mg/week) groups. Using 28 and 49 mg/week as threshold points, we assessed the association of *CYP2C9* and *VKORC1* haplotypes with warfarin sensitivity and resistance. Furthermore, the linkage disequilibrium (LD) between different alleles of *CYP2C9* and *VKORC1* was also calculated by using SHEsis web-based platform.

Resolution of warfarin enantiomers by HPLC

The (R)- and (S)-enantiomers of warfarin and 7-hydroxywarfarin in human plasma were retrieved by solid-phase extraction using an Oasis HLB cartridge. These analytes along with the internal standard (p-chlorowarfarin) were separated on a chiral CD-Ph column by using a mobile phase of 0.5% KH_2PO_4 (pH 3.5)–methanol (41:59, v/v) at a flow rate of 0.5 ml/min and were detected at λ_{max} of 305 nm [23]. The sensitivity of detection of each analyte was 2.5 ng/ml. The inter- and intra-assay precisions were 10.6 and 10.2%, respectively. The accuracies were within 6.3% in the linear range.

Table 1. Genetic analysis.

SNP	Primers	A	x	RE	RE site
<i>CYP2C9*2</i>	f-TACAAATACAATGAAAATATCATG r-CTAACAACCAGACTCATAATG	60°C	35	Avall	-
<i>CYP2C9*3</i>	f-AATAATAATATGCACGAGGTCCAGAGATGC f-AATAATAATATGCACGAG GTCCAGAGGTAC r-GATACTATGAATTTGGGACTTC	50°C	35	Nsil KpnI	+
<i>CYP2C9*8</i>	f-TACAATACAATGAAAATATCATG	58°C	35	Acil	-
<i>CYP2C9*13</i>	r-CTAACAACCAGGACTCATAATG			MspI	+
<i>VKORC1*3</i>	f-TACAATACAATGAAAATATCATG r-CTAACAACCAGGACTCATAATG	64°C	30	Acil	+
<i>VKORC1*4</i>	f-GCA TAATGACGGAATACAGAGGAGGC r-GGTAGAGACAGGCTTTCACCATGT	61°C	30	Bfal	+
<i>VKORC1 D36Y</i>	f-AACCTGGAGATAATGGGCAGCA r-ACACCGATCCCAGACTCCAGAATA	63°C	30	RsaI	+
<i>VKORC1-1639 G>A</i>	f-ATCCCTCTGGGAAGTCAAGC r-CACCTTCAACCTCTCCATCC	60°C	35	NciI	-
<i>CYP4F2 V433M</i>	f-CGGAACCTGGACCATCTACA r-CC TACTCTCCACAGGCATTA	60°C	35	PvuII	-
<i>GGCX</i>	f-GCTGGTGTCTACTGCCCGC r-GTACTCTCCACAGCCCTACAAA	59°C	35	BstUI	-

+: Creation of restriction site due to the polymorphic variant; -: Abolition of restriction site due to the polymorphic variant; x: Number of cycle; A: Annealing temperature; RE: Restriction enzyme.

Determination of vitamin K (K1, K2, K3) by HPLC

Phenomenex Strata-X 33 μ (30 mg/ml) polymeric reverse-phase cartridges were used for solid-phase extraction of vitamins K1, K2 and K3 from the plasma samples of the patients. These cartridges were pre-conditioned with 350 μ l 100% methanol followed by 350 μ l double-distilled water (DDW). Plasma samples (250 μ l) were added to the preconditioned cartridges. After the addition of plasma, the cartridge was then washed with 350 μ l of DDW, 350 μ l of DDW + methanol (50:50) and 350 μ l of methanol and DDW (20:80) in three different steps. Finally, the vitamins K1, K2 and K3, which were retained in the stationary phase of the cartridges were eluted with 2 $\times$ 0.5 ml of 100% diethyl ether. Furthermore, the eluent was dried in a water bath at 60°C in sample evaporator. The dried samples were reconstituted with 120 μ l of vitamins K1, K2 standards (30 ng/ml), xanthophyll internal standard (60 ng/ml) and methanol (30 ng/ml), and 100 μ l was injected into C18 reverse-phase HPLC column.

Acetonitrile and methanol were used as mobile phase in 1:1 ratio at a flow rate of 1.3 ml/min. The detection and identification of the analytes was performed by using photodiode array detector using 280 nm for

vitamins and 450 nm for internal standard. Separation was performed on a C18 column (Enable C18 H 250 $\times$ 4.6 mm, Spincotech Pvt Ltd, Chennai, India). The retention times (minutes) for vitamins K1, K2 and K3 were 19.5, 9.6 and 2.5, respectively. The detection limits were 1.3–6.2 ng/20 μ l injected samples. The extraction recoveries estimated to be between 94 and 95.6%. The inter- and intra-assay precisions varied from 9.6 to 10.7%, respectively.

Development of ANN model

An ANN was developed from the data set of derivation cohort. The variables such as age, gender, BMI, warfarin dose, *CYP2C9*2*, *CYP2C9*3*, *CYP2C9*8*, *CYP2C9*13*, *VKORC1*3*, *VKORC1*4*, *VKORC1-1639 G>A*, *VKORC1 D36Y*, *CYP4F2 V433M*, *GGCX* G8016A, plasma vitamins K1, K2, K3 and TSH levels were included as the inputs and warfarin dose (mg/week) as the output.

Among all the transfer functions, hyperbolic tangent worked efficiently when the output is a continuous variable. The number of hidden layers and nodes were adjusted until the least sum of error was obtained. The preprocessing of data was done by the normalization where in the data in a column will be

Table 2. Genotype distribution in the studied population.										
SNP	Derivation cohort (n = 157)				P _{HWE}	Validation cohort (n = 83)				P _{HWE}
	0	1	2	VAF		0	1	2	VAF	
CYP2C9*2	119	35	1	11.94	0.73	63	19	0	11.58	0.22
CYP2C9*3	131	25	1	8.59	1.00	69	14	0	8.43	0.51
CYP2C9*8	150	7	0	2.22	1.00	79	4	0	2.41	1.00
VKORC1*3	99	56	2	19.11	0.23	65	18	0	10.84	0.53
VKORC1*4	127	26	4	10.82	0.28	74	9	0	5.42	1.00
VKORC1 -1639 G>A	108	41	2	14.90	0.80	68	13	2	10.24	0.53
CYP4F2	43	80	34	47.13	0.95	21	37	25	52.41	0.68
GGCX	115	41	1	13.69	0.41	63	18	2	13.25	0.59
SNP	Clinical dosing group (n = 107)				P _{HWE}	Genetically guided group (n = 100)				P _{HWE}
	0	1	2	VAF		0	1	2	VAF	
CYP2C9*2	84	22	1	11.22	1.00	80	20	0	10.00	0.54
CYP2C9*3	88	18	1	9.35	1.00	85	15	0	7.50	0.52
CYP2C9*8	106	1	0	0.46	1.00	96	4	0	2.00	1.00
VKORC1*3	73	33	1	16.36	0.38	60	37	3	21.50	0.75
VKORC1*4	88	16	3	10.28	0.09	82	17	1	9.50	1.00
VKORC1 -1639 G>A	84	22	1	11.22	1.00	73	25	2	14.50	1.00
CYP4F2	24	61	22	49.07	0.30	32	50	18	43.00	1.00
GGCX	74	32	1	15.89	0.65	82	17	1	9.50	1.00
0: Wild genotype; 1: Heterozygous genotype; 2: Homozygous genotype; HWE: Hardy–Weinberg equilibrium; P: Calculated probability; VAF: Variant allele frequency.										

divided by maximum number in order to maintain all the values in the range of 0 to 1. The data of derivation cohort were used for training the ANN. The data of derivation and validation cohorts were used for validation of the model. To generate the ANN model, the ‘Visual Gene Developer’ version 1.4 was used [24]. The experimental and the predicted data (derivation cohort) were correlated by using the Spearman correlation coefficient, and the square of which gives the percentage variability of warfarin pharmacokinetics.

Testing the clinical utility of ANN model

A total of 207 patients randomized into two groups with Group A dose adjusted based on ANN algorithm (50 men and 50 women) and Group B dosage adjustment based on clinical parameters (61 men and 46 women). From the inpatient numbers, random numbers were generated by computer and were assigned to either Group A or Group B. The mean ages of these groups were 37.95 ± 14.99 versus 39.05 ± 15.42 years (p = 0.60). The time schedule for genetic analysis and ANN analyses was 2–3 days. During this period, the dose in Group A was determined based on age, gender and BMI. Later, ANN-based dose was initiated. In Group B, the initial dose was 5 mg/day, which was

lowered if INR more than 3.5 or increased if INR less than 2.0 until the INR reaches therapeutic target, in other words, 2.0–3.5. The patients were followed up for the duration of 90 days and INR values were recorded on six intervals (day 0, day 3, day 5, day 21, day 60, day 90). The time to reach first therapeutic INR and out-of-range INRs (INR >3.5 and <2.0) was calculated based on INR data collected at different intervals following the initiation of warfarin therapy. The adverse drug reactions in both the groups were of monitored during this period. The time to reach first therapeutic INR, out-of-range INRs and rate of adverse reactions were used as indices to demonstrate clinical utility of the ANN model.

Statistical analysis

Student t-test was performed to ascertain the association of hypothyroidism versus euthyroidism influence on warfarin dose. Spearman rank correlation was used to assess the agreement between actual warfarin dose and predicted dose where in ‘r’ represents correlation coefficient. The ‘r²’ value is an index of percent variability explained by that model. Bland and Altman analysis was carried out to assess the mean absolute error (MAE). The Fisher exact test was performed by

computing the data in 2×2 contingency table based on the variable distribution (yes/no) between the two groups. Odds ratios (ORs) and 95% CIs were obtained by this analysis.

Results

As shown in Table 2, the frequencies of all the polymorphisms in derivation as well as validation cohort were found to be in accordance with the Hardy–Weinberg equilibrium (p_{HWE} 0.52–1.00).

As shown in Table 3, univariate analysis revealed gender, *CYP2C9**3 and *CYP2C9**8 as the statistically significant determinants of warfarin dose. As shown in Figure 1A, the developed ANN model showed excellent predictability ($R^2 = 0.959$), thus explaining 95.9% of variability. The MAE in the predictability was -1.97 mg/week. The MAE values across warfarin sensitive (<28 mg/week), warfarin intermediate (28–49 mg/week), warfarin resistant (>49 mg/week) groups were in acceptable limits (Figure 1B). The developed ANN model was applied to calculate both the overestimation rate and underestimation rate. As shown in Figure 2A, this model predicted the need for higher warfarin dose in 74.5% patients whose INR was less than 2.0, and the need for lower warfarin dose in 83.3% patients whose INR was more than 3.5. The current model was unable to predict higher dose and lower dose than the actual in 13 patients with INR less than 2.0 and two patients with INR more than 3.5 ($n = 2$), respectively. In rest of the patients, in other words, 93.75%, the ANN model performed well.

Furthermore, hypothyroidism was observed to be associated with higher-dose requirements of warfarin. The current ANN model performed well both in euthyroid and hypothyroid states (Figure 2B).

SHEsis analysis revealed moderate-to-strong LD (D' : 0.59–1.00) among *CYP2C9**2, *CYP2C9**3 and *CYP2C9**8. *VKORC1* -1639 G>A and *VKORC1**3 are in strong LD (D' : 0.60). No LD was observed between *VKORC1* -1639 G>A and *VKORC1**4 (D' : 0.07) while mild LD was observed between *VKORC1**3 and *VKORC1**4. Six *CYP2C9* haplotypes and eight *VKORC1* haplotypes were found to be predominant in our population.

As shown in Figure 3, we applied ANN simulations to predict the warfarin dose based on *CYP2C9* and *VKORC1* haplotypes. Based on these simulations, we classified the patients into warfarin sensitive, intermediate and resistant groups. Warfarin sensitivity was associated with CCG and TCG haplotypes of *CYP2C9* (*CYP2C9**2 and *CYP2C9**3 variants) and *VKORC1* AAC haplotype (*VKORC1*-1639 G<A).

The applicability of ANN algorithm in an independent data set

To test the clinical utility of ANN algorithm, we have randomly assigned genetically guided dosing to an independent cohort ($n = 207$). We observed significant reduction in the out-of-range INRs in the genetic testing group (OR: 0.49; 95% CI: 0.30–0.79; $p = 0.003$). The genetically guided group exhibited reduction in the rate of adverse drug reactions in comparison to

Table 3. Individual contribution of variables with warfarin dose.

Variable	Correlation coefficient	p-value
Age	-0.18	NS
Gender	0.22	<0.05
BMI (kg/m ²)	-0.09	NS
<i>CYP2C9</i> *2	0.13	NS
<i>CYP2C9</i> *3	-0.23	<0.05
<i>VKORC1</i> *3	-0.12	NS
<i>VKORC1</i> *4	-0.04	NS
<i>VKORC1</i> -1639 G>A	-0.06	NS
<i>CYP4F2</i>	-0.02	NS
<i>GGCX</i>	0.11	NS
<i>CYP2C9</i> *8	0.28	<0.01
Vitamin K1	0.001	NS
Vitamin K2	-0.14	NS
Vitamin K3	0.06	NS
Hypothyroidism	0.09	NS

NS: Not statistically significant.

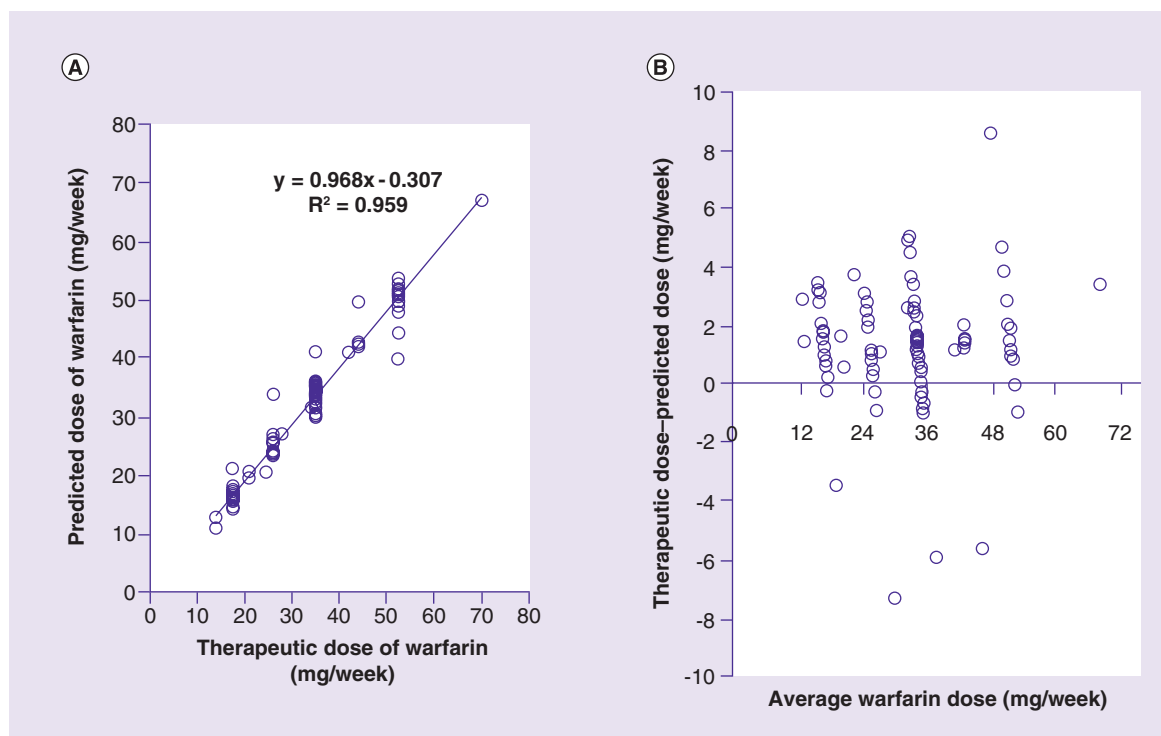


Figure 1. The prediction accuracy of the artificial neural network-based pharmacogenomic algorithm. (A) The predictability of artificial neural network-based pharmacogenomic algorithm was demonstrated by using the therapeutic dose data of derivation cohort ($R^2 = 0.959$). (B) The Bland–Altman analysis between therapeutic and predicted data showing good agreement (bias: ± 6 mg/week).

clinical dosing group (OR: 0.00; 95% CI: 0.00–0.52; $p = 0.002$). The adverse drug reactions observed in clinical dosing group were: hemorrhage and gangrene of limbs ($n = 3$), bleeding events ($n = 3$), skin necrosis ($n = 1$) and death ($n = 1$). None of the subjects in the genetically guided group had any adverse drug reactions. The mean time to achieve first therapeutic INR was lesser in the genetically guided group compared with clinical dosing group (29.7 ± 32.7 vs 47.3 ± 34.1 days; $p < 0.0003$). The genetically guided group attained first therapeutic INR 6.73-folds (95% CI: 2.17–22.31; $p < 0.0001$) faster than clinical dosing group. Figure 4 represents percentage of INRs in therapeutic range across different time points. The genetically guided dosing showed better clinical utility compared with clinical dosing in terms of percentage of INRs in therapeutic range.

S-warfarin/7-hydroxywarfarin ratio according to *CYP2C9* genotypes

As shown in Figure 5, the ratio between S-warfarin and 7-hydroxywarfarin was found to be lower in *CYP2C9*1* indicating the extensive metabolizer phenotype. This ratio was observed to be higher in *CYP2C9*3* compared with *CYP2C9*2*. The compound heterozygous genotype of *CYP2C9* ($*2/*3$) was found to be associated with the highest ratio

indicating the poor metabolizer phenotype. Warfarin metabolism was found to be rapid in younger individuals. The metabolic rate was found to reduce with increase in BMI in men while the metabolic rate was found to increase with increase in BMI in women.

Discussion

The concept of applying ANNs for the development of pharmacogenomic algorithms is naïve. Till now, four ANN models were developed taking to account only the clinical and the demographical variables for warfarin dose prediction [25–28]. Only one study has taken genetic variables into account and it showed better predictability with an average absolute error of 5.7 mg/week [29]. The current algorithm incorporated additional genetic variants such as *GGCX* and *CYP2C9*3* along with vitamin K and thyroid status and hence exhibiting higher predictability than any existing algorithms. Furthermore, being a dynamic algorithm, its size, dimension and the error rate can be minimized continuously.

Till date, several pharmacogenomic algorithms were proposed to optimize warfarin dose. Gage *et al.* were able to explain 39% variability in warfarin dosing based on age, body surface area, *CYP2C9*2*, *CYP2C9*3*, gender, race, amiodarone and simvastatin levels [16]. Sconce *et al.* explained 55% variability using

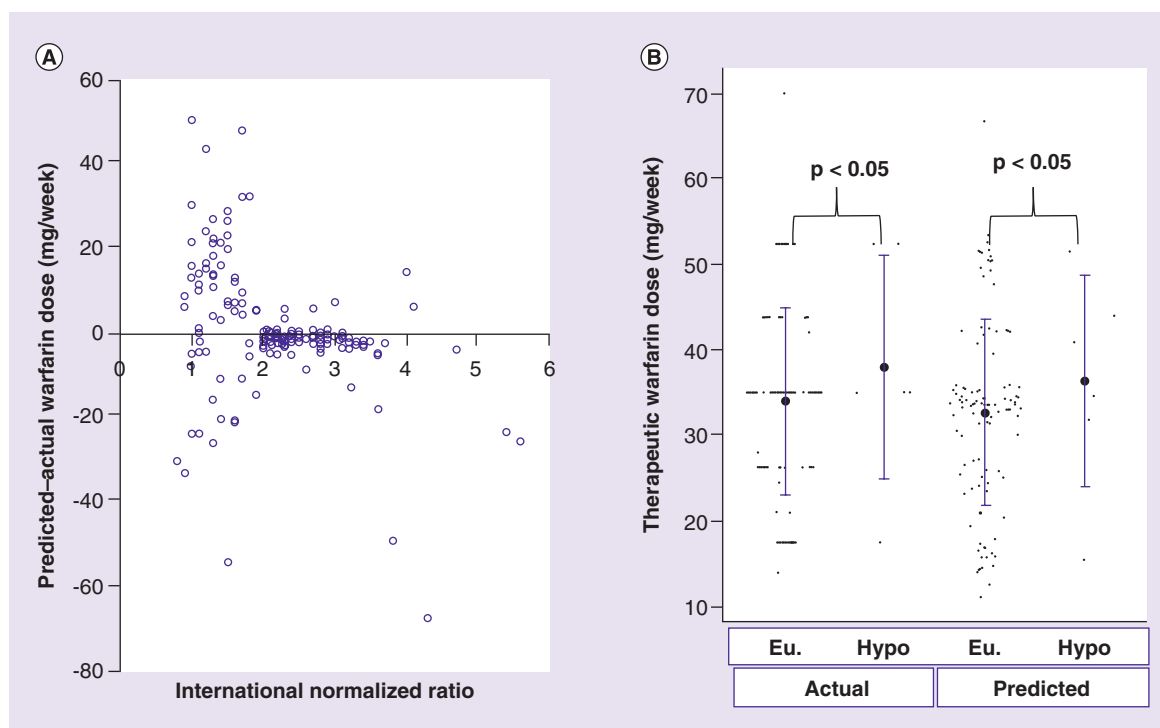


Figure 2. Validation of the artificial neural network-based pharmacogenomic algorithm. (A) The artificial neural network-based pharmacogenomic algorithm was applied to all the patients across different international normalized ratios (INRs). In INR <2.0 group, this algorithm predicted the need for higher dose in 74.5% patients while it predicted the need for a lower dose in 83.3% patients whose INR >3.5. (B) Hypothyroidism was shown to be associated with higher dose of warfarin, which was also predicted by the artificial neural network algorithm.

age, *CYP2C9**2, *CYP2C9**3, *VKORC1* -1639G<A and height [30]. Zhu *et al.* explained 61% variability in warfarin dose by including age, gender, weight, *CYP2C9**2, *CYP2C9**3 and *VKORC1* -1639G<A [14]. Wadelius *et al.* were able to explain 59% variability by including *CYP2C9**2, *CYP2C9**3 and *VKORC1* -1639G<A along with age, gender and drugs that increase INR [15]. In order to make a working algorithm suitable for all the worldwide populations, International Warfarin Pharmacogenomic Consortium (IWPC) developed an algorithm using age, height, weight, *CYP2C9**2, *CYP2C9**3 and *VKORC1*

-1639G<A, race, enzyme inducer and amiodarone as predictors [31]. This model explained 66% variability. IWPC algorithm was not able to account for majority of Indians as they have low frequency of *CYP2C9**2 and *CYP2C9**3 and high frequency of *VKORC1**3 and *VKORC1**4. To address this issue, we have developed a population-specific algorithm that accounts for 44.9% variability [18]. Further by including *CYP2C9**8, *CYP4F2*, *GGCX* and thyroid status, we have developed an expanded genetic model which explained 61% variability in warfarin dosage requirement [19]. The current ANN model explains

<i>CYP2C9/VKORC1</i>	GGC	AGC	GAC	GGT	AAC	GAT	AGT	AAT
CAG	8.1	7	5.1	3.7	3.7	6	3	3
TAG	7.7	7	4.3	7.4	4.4	6	5.4	4
CCG	5	5	2	5	1.5	6	4.2	5.8
TCG	6.3	6	3	7	2	4.5	6	5
CAA	7.6	6.4	9	3.6	9	8	2	6
TAA	9	8.5	8	7	8	9	6	8

Figure 3. Influence of pharmacokinetic and pharmacodynamic genetic variants on the warfarin dose (mg/day).

The green color background suggests warfarin sensitivity and the red color background suggests warfarin resistance. The yellow and orange colors are representative of warfarin intermediate group. *CYP2C9* haplotype covers *CYP2C9**2 (C>T), *3 (A>C) and *8 (G>A) alleles; *VKORC1* haplotype covers *VKORC1*-1639 G>A, *3 (G>A) and *4 (C>T) alleles.

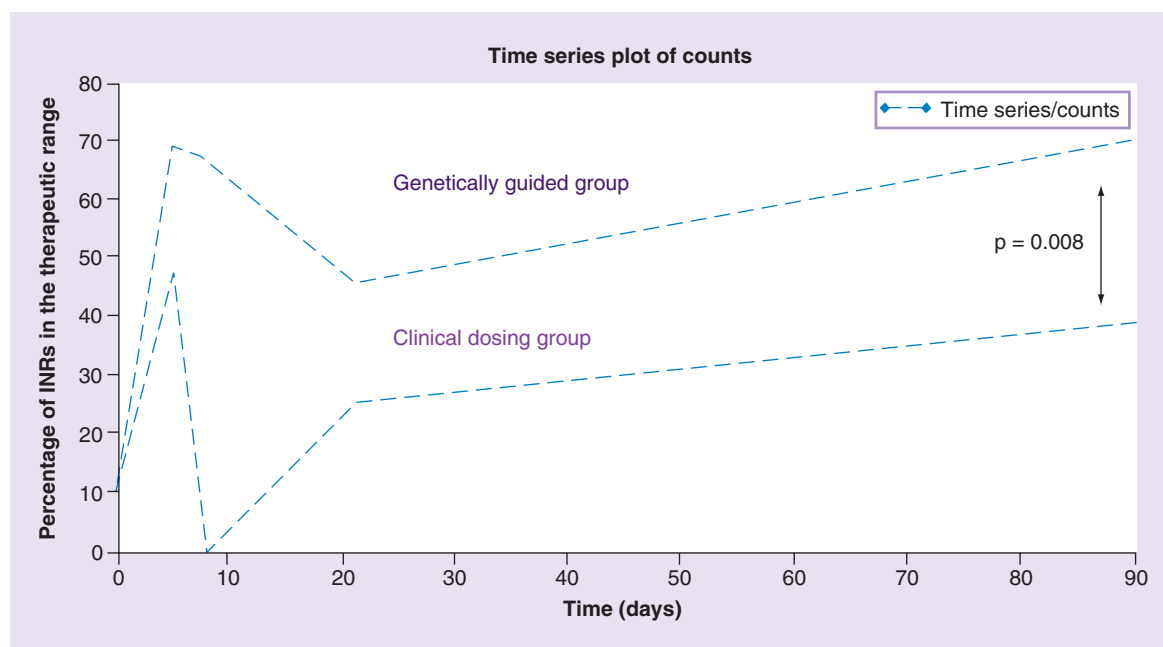


Figure 4. The performance of genetically guided dosing versus clinical dosing. The patients were randomized into two groups: (A) genetically guided (n = 100) and (B) clinical dosing (n = 107). INR values were measured on six intervals, in other words, day 0, day 3, day 9, day 21, day 60 and day 90. INR: International normalized ratio.

93.5% variability as it can explore even nonlinear interactions between genetic variants.

In the studied population, none of the subjects had *CYP2C9*13* variant. SHEsis-based haplotype analysis showed strong linkage disequilibrium between *CYP2C9*3* and *CYP2C9*8*; and *CYP2C9*2* and

*CYP2C9*8* suggesting that these three variants are part of a haplotype block and their effects will be haplotype-based rather than genotype-based.

Our results are consistent with the existing literature in attributing the poor metabolizer phenotype to *CYP2C9*2* and *CYP2C9*3* the higher S-warfarin/7-OH

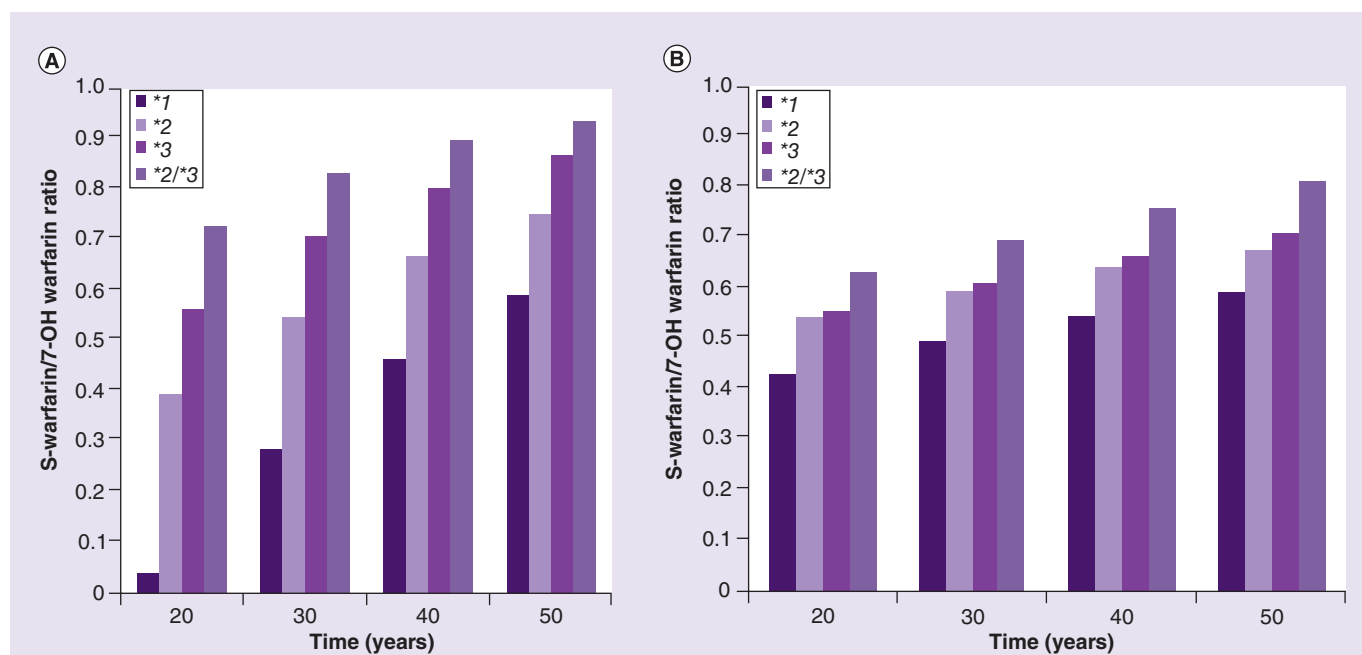


Figure 5. Variations in S-warfarin hydroxylation based on *CYP2C9* genotypes. In men (A), the S-warfarin hydroxylation was found to be more rapid than women (B). The *CYP2C9*1* (wild genotype) is excessive metabolizer depicting lower S-warfarin/7-OH warfarin ratio. The hydroxylation was found to decrease in the following order: *2<*3<*2/*3.

warfarin ratio [32,33]. Apart from *CYP2C9* polymorphisms, Phase II enzymes such as UDP-glucuronosyl transferases that cause glucuronidation of R- and S-hydroxyl warfarin [34]; Phase I enzymes such as *CYP2C19* that carry hydroxylation of R- and S-forms of warfarin [35]; drug–drug interactions [36,37] also reported to influence warfarin pharmacokinetics.

The developed ANN model had lower rates of underestimation and overestimation. Application of ANN-based algorithm in the current study reduced the out-of-range of INRs by 49% and reduced the time to reach first therapeutic INR by 6.6-folds with absolutely no adverse drug reactions in the genetically guided group. Our results are in agreement with Pirmohamed *et al.*, who demonstrated increased time in therapeutic INR range in pharmacogenetically guided dosing group [38]. However, Kimmel *et al.* observed no benefit in anticoagulation therapy following genotype-guided dosing [39]. The lack of clinical benefit in Kimmel *et al.* study could be due to inclusion of only *CYP2C9* and *VKORC1* genetic polymorphisms in the algorithm. Probably, an expanded genetic model might be more beneficial in ensuring effective and safe anticoagulation.

The limitations of ANN model are: if learning and momentum coefficient are very low, the performance of the model will go down. If learning rate and momentum coefficient are too high or number of hidden layers are too many, there is over fitting of data. Hence, balancing these training parameters to a optimal value is the key component that ensures success of the ANN model. The second limitation of the current model is less representation of elderly subjects with atrial fibrillation. Future studies incorporating more number of elderly subjects

with atrial fibrillation will improve its applicability across different age groups. In the current study, we have performed a two-tier validation of our data: first, by comparing the predicted dose with therapeutic dose and second, by testing the predictability of the model in patients with out of range INRs. Furthermore, the clinical utility of this model was demonstrated in various aspects such as agreement between therapeutic dose and predicted dose in euthyroid and hypothyroid, the benefit of genetically guided dosing in minimizing out of range INRs, decreasing the time to reach first therapeutic INR and reducing rate of adverse drug reactions.

Conclusion

To conclude the algorithm based on ANN showed 92.5% accuracy with 5.8% of the underestimation rate and 0.4% overestimation rate. The developed model is highly precise with MAE of 1 mg/week. Performance of this model was excellent in both euthyroid and hypothyroid status. This ANN algorithm reduced out-of-range INRs, minimized time to reach first therapeutic INR and reduced rate of adverse drug reactions.

Future perspective

This ANN model can be updated periodically with data from various geographical areas and ethnic groups, thus making its application versatile in predicting optimal dose of warfarin. Such models might ensure therapeutic efficacy, reduce the burden of adverse drug reactions and reduce hospitalization events.

Financial & competing interests disclosure

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Executive summary

Background

- In order to develop more precise pharmacogenomic algorithm for warfarin dose optimization and to incorporate all the possible demographic, genetic, nutritional variables into the model, we have applied artificial neural networks (ANNs).

Materials & methods

- An ANN algorithm was developed by using age, gender, BMI, plasma vitamin K levels, thyroid status and ten genetic variables as the inputs and therapeutic warfarin dose as the output.
- Hyperbolic tangent function was used to build an ANN architecture.

Results & conclusion

- The ANN model explained 93.5% variability in warfarin dosing.
- This model predicted warfarin dose accurately in 74.5% patients whose international normalized ratio (INR) was less than 2.0 and in 83.3% patients whose INR was more than 3.5.
- This algorithm reduced the out-of-range INRs (OR: 0.49; 95% CI: 0.30–0.79; $p = 0.003$), the rate of adverse drug reactions (OR: 0.00; 95% CI: 0.00–1.21; $p = 0.06$) and time to reach first therapeutic INR (OR: 6.73; 95% CI: 2.17–22.31; $p < 0.0001$).
- This algorithm was found to be applicable in both euthyroid and hypothyroid status.
- S-warfarin/7-hydroxywarfarin ratio was found to increase in subjects with *CYP2C9**2 and *CYP2C9**3 justifying the warfarin sensitivity attributed to these variants.
- An application of ANN for warfarin dosing improves predictability and provides safe and effective dosing.

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Ethical conduct of research

The authors state that they have obtained appropriate institutional review board approval and have followed the principles outlined in the Declaration of Helsinki for all human or animal experimental investigations. In addition, for investigations involving human subjects, informed consent has been obtained from the participants involved.

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